

3 $\forall a, b, k, k \neq 0$

$$\int_a^b s\left(\frac{x}{k}\right) dx = k \int_a^b s(x) dx$$

proof:

① $a < b, k > 0$

② $a < b, k < 0$

③ $a > b, k \neq 0$

we prove ② first.

②: let $P = \{x_0, x_1, \dots, x_n\}$ be partition of $[a, b]$ such that s is constant on the open subintervals of P .

let $s(x) = s_i$ if $x \in (x_{i-1}, x_i)$

let $t(x) = s\left(\frac{x}{k}\right)$, if $kb \leq x \leq ka$

$\therefore t(x) = s_i$ if $x \in (kx_{i-1}, kx_i)$

$\therefore P' = \{kx_0, \dots, kx_{n-1}\}$ is

partition of $[kb, ka]$ and t is constant on the open subintervals of P' .

$$\therefore \int_a^b t(x) dx$$

$$= \sum_{i=1}^n s_i (kx_i - kx_{i-1})$$

$$= -k \sum_{i=1}^n s_i (x_i - x_{i-1})$$

$$= -k \int_a^b s(x) dx$$

$$\therefore \int_a^b s\left(\frac{x}{k}\right) dx$$

$$= - \int_a^b t(x) dx$$

$$= k \int_a^b s(x) dx$$

② is proved.

① can be similarly proved.

③: If $a > b, k \neq 0$

$$\int_a^b s\left(\frac{x}{k}\right) dx = - \int_b^a s\left(\frac{x}{k}\right) dx$$

$$= -k \int_b^a s(x) dx$$

$$= k \int_a^b s(x) dx$$

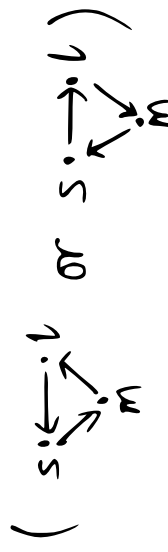
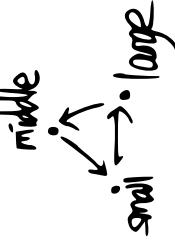
□

Integral of step function

1. $\int_a^c s(x) dx + \int_c^b s(x) dx = \int_a^b s(x) dx$, if $a \leq c \leq b$

$$\Rightarrow \int_a^c s(x) dx + \int_c^b s(x) dx = \int_a^b s(x) dx = 0$$

$$\int_c^a s(x) dx + \int_a^b s(x) dx = 0$$



$$\Rightarrow = 0$$

2. $\forall a, b, c, \int_a^c s(x) dx + \int_c^b s(x) dx + \int_b^a s(x) dx = 0$



if a is middle value, rotate the graph,

